

Process Mining Did Not Detect the Faults

Measured results from an application to HVAC, and the protocol that refuted the hypothesis behind it

Yassh Singh

Thompson Rivers University, Kamloops, BC, Canada

Abstract. We report a negative result and the protocol that produced it. Discovering Petri nets from fault-free operation of three building HVAC systems, we tested whether conformance against them detects seeded faults. Over all 73 seeded scenarios no fault is detected only by the two conformance channels; removing them changes no scorecard and lowers one system’s false-alarm rate from six days in 96 to one, every row out-of-sample. Three further measurements support that reading: a measured circularity bound, 1.78% against 5.24% recall at one threshold; a reversal probe showing that two of the three discovered nets carry no ordering information in their fitness; alignment-based support that adds nothing over counting; and a pre-registered transfer test that never had attainable power. We report the protocol — pre-registration with binding falsifiers, adversarial recomputation, tests that pin disclosures — as what made the null checkable.

Keywords: Process mining · Conformance checking · Event abstraction · Negative results · Reproducibility · Cyber-physical systems.

1 Introduction: an empty quadrant, and what we found in it

Process mining has spread from business processes into cyber-physical and sensor-driven domains, and the spread has been reported as a success. One domain of that kind is missing. Building heating, ventilation and air-conditioning (HVAC) operation is a continuous-flow, standard-governed, heavily instrumented process, and it does not appear: the one systematic review of process mining on sensor data whose scope could have captured it records no building HVAC application [6], and a survey of 211 AI-based HVAC fault-detection studies has no process-mining entry at any leaf of its method taxonomy [4].

The quadrant looked worth filling for a specific reason. The largest empirical characterisation of HVAC faults, across more than 60,000 pieces of equipment in 317 buildings, catalogues the ten most common air-handler faults [7], and on our reading seven of them are behavioural rather than component-level: schedules at the wrong hours, overrides never released, resets skipped, economiser sequences executed out of order. These are deviations in the ordering of operating states, and the threshold alarms and rule libraries buildings actually run see a wrong

order only if someone wrote a rule for it in advance. Our hypothesis was the obvious one: a model of the order in which a building moves between states should have an advantage on exactly the faults that dominate the field. We pre-registered it with the observation that would refute it, and built the pipeline to test it.

It was refuted, and this paper reports the refutation with the measurements that explain it. Discovery, conformance and detection were built before any result below was known, and the falsifier for the founding hypothesis — that a hand-written rule would out-cover the discovered models on the same scenarios — was written down before the rule existed. The field has almost no published record of that; the one precedent we know of, in another cyber-physical domain, is a preprint reporting that naïve process mining on network event logs was ineffective for intrusion detection [23]. Ours is accompanied by artefacts from which every number regenerates. The HVAC domain is the setting; the results are about process mining.

2 Related work

Conformance as an anomaly detector. Scoring traces against a discovered model to flag anomalous ones is an established line [3], on conformance foundations laid by Rozinat and van der Aalst [17, 1]. The line then moved: neural sequence models trained on event logs are reported to outperform conformance-based baselines on standard corpora [15]. Our null belongs with that migration rather than against it. What the neural line has not done, and Section 5 attempts, is to hold the comparison fixed at the level of the individual channel inside one detector on one dataset, where the question — is the conformance signal ever the thing that catches a fault? — has a yes-or-no answer per scenario that can be read off a committed artefact.

Event abstraction and leakage. Turning sensor streams into activity sequences is the binding step for any sensor-domain application; we follow Janssen and colleagues [11] within the taxonomy of van Zelst and colleagues [22]. The one decision in that step with a measurable consequence is a leakage decision, in the sense Weytjens and De Weerdts catalogue for process-mining experiments [20]: Section 3 measures what a discovered model loses when it is no longer allowed to recognise the analyst’s own rules.

Precision, and why we did not use it. Precision metrics [14, 19] are ill-posed on logs without negative examples [5], which sensor logs never have; we replaced them with a direct probe (Section 4).

Protocol. Pitsch and colleagues argue that process-mining research should adopt explicit hypothesis testing [16]; the pre-registration and binding falsifiers behind this paper are one attempt to practise it. Evaluation uses the public, DOI-citable LBNL benchmark family [10]; the domain’s own methods and their reproducibility gap are surveyed elsewhere [4].

3 Event abstraction, the alphabet wall, and a measured circularity bound

3.1 The abstraction

HVAC operation offers no natural case boundary: a building runs continuously, with concurrent equipment at several granularities and signals that are sampled rather than emitted. We take the calendar day as the case, the shortest unit over which the control sequences of interest — start-up, occupancy, economiser transitions, night cycling, shutdown — complete. Streams become events by control-sequence rules grounded in ASHRAE Guideline 36 [11]. The canonical alphabet has ten activities in five complementary pairs (system started and stopped, night cycle started and ended, heating active and inactive, cooling active and inactive, economiser window entered and exited);¹ the single-duct air handler’s configuration yields six of them, the two fan-powered units all ten. Because every activity has an inverse that also occurs many times a day, the traces are loop-heavy — the calibrated healthy band for cooling-active occurrences on the series unit runs from 4 to 44 in a single day — and the alphabet is constructed, not emitted: each activity exists because a rule decided that a sustained threshold crossing counts as a state change. That is ordinary for sensor-domain process mining [22], and it is the opening through which the analyst’s judgement enters the model.

3.2 The alphabet wall

The events are partitioned into two alphabets that never mix. *State events* are neutral facts about what the equipment did and are the only events admitted to discovery. *Fault-signature events* already encode engineering suspicion — a valve commanded open while the coil it feeds does nothing, a supply temperature contradicting the unit’s claimed mode — and are withheld from discovery entirely, though they remain available to the detection side. The partition exists because of circularity, the process-mining form of leakage [20]: a model discovered over an alphabet that already names faults will later “detect” them, but what it has recognised is the analyst’s own rule, re-encoded as a transition and read back out through alignment cost. The literature warns about this in the abstract; we are not aware of a prior quantification on a sensor-derived log.

3.3 The bound

Quantifying it needs one comparison, and we state exactly what the comparison is. The unpartitioned number comes from the committed artefact of the pre-partition prototype pipeline; the partitioned number from this pipeline’s artefact on the same single-duct system. The two share the alignment threshold, the 248-day training and 87-day holdout split, the 4,540 scored fault days, and — the

¹ `outputs/grammar_results.json`, `canonical_alphabet` and `event_days`.

check that the comparison is not confounded elsewhere — an identical rules-channel output of 3,134 true positives; they do not share a code version, and a controlled same-code re-run is future work. The deployed scorecard, two code generations later, gives the same reading: 86 of 4,891 evaluable fault days strictly worse than the worst fault-free holdout day (the thresholded channel flags none), 1.76%.² Under that comparison the model channel’s apparent recall rises from 1.78% to 5.24% — 81 detected fault days against 238 — when the discovered net may contain the activities that name the faults.³ The difference, 3.5 percentage points, is a measured bound on model self-confirmation for this system and this alphabet. The two artefacts also agree on every combined-detector number — 3,152 true positives and 5 false positives in both — so the partition changed the deployed detector’s output by exactly zero days: every model-channel detection the leak had supplied was already caught by the rules channel, which is the redundancy the ablation of Section 5 later measured directly. It is a bound on one system and on inflated recall only; the design is generic.

3.4 Pinning the partition

A partition enforced by convention decays, so the separation is held by a tested invariant: injecting fault-signature events into a log must not change any day’s fitness against a discovered net, and the test fails at build time if a future edit lets signature events reach the discovery path.

4 Discovery at three strata, and the reversal probe

Petri nets are discovered from the state-event log (Table 1) with the inductive miner [13] in pm4py [2], at three granularities: a *device* stratum modelling one actuator’s lifecycle with one case per device per day, a *unit* stratum modelling the air handler’s daily cycle, and a cross-system grammar mapping each system’s alphabet onto the canonical ten activities so that a net discovered on one building can be replayed against another’s log. All discovery uses fault-free data only.

4.1 The reversal probe

Whether such a net encodes temporal order, or merely licenses a set of activities in any arrangement, is the question precision metrics answer least reliably on logs with no negative examples [19, 5], so we use a direct probe [8]: reverse every trace, re-score it against the same net, and report the drop in mean fitness. A net that constrains order cannot fit a reversed log as well as the original; a net whose fitness is a function of the activity multiset will not notice. The three nets

² `outputs/benchmark_v6_sdahu.json, model_days.`

³ Partitioned run: `outputs/benchmark_v3.json`; unpartitioned predecessor: `outputs/benchmark_v2_processheal_v1.json`. Both report threshold 0.333, `train_days` 248, `holdout_days` 87, `pos_days` 4,540; the `structure` channel’s true positives are 81 and 238.

Table 1. The three unit-stratum logs and nets. Activities, event days and variants are counted over the fault-free year; “in net” counts the activities the miner kept as labelled transitions; net sizes are places / transitions / silent transitions; fitness is mean alignment fitness on the holdout days ($n = 87 / 96 / 96$), forwards and on reversed traces (Section 4).

System	Act. (in net)	Days	Variants	Net p/t/ τ	Fitness	Reversed
Single-duct AHU	6 (6)	335	102	16/17/11	0.8861	0.5526
Parallel fan-powered	10 (7)	365	343	15/18/11	0.9021	0.9026
Series fan-powered	10 (8)	365	364	15/17/9	0.9533	0.9533

separate cleanly.⁴ The single-duct net falls from 0.8861 to 0.5526: it encodes order. The series fan-powered net scores 0.9533 forwards and 0.9533 backwards, and the parallel net moves from 0.9021 to 0.9026, five ten-thousandths in the wrong direction. Two of the three nets carry no ordering information in their fitness; their high absolute fitness states which activities occurred, not when. Its limit: it tests whether *fitness* responds to order, not whether the net contains order-encoding structure the log never visits.

4.2 The net’s reading added nothing over counting

A further measurement belongs here although it was computed for the test in Section 6. For each system we counted the fault-free occupied days on which a named activity is replayed as a synchronous move against the discovered unit net. On all three systems sealed at the time the count equalled the number of days on which the activity simply appears in the raw log: 0 against 0, 166 against 166, and 357 against 357.⁵ At the unit stratum the alignment added nothing that counting would not have given, exactly. The device stratum, whose net is pooled across each building’s four terminal units, disagrees with the unit stratum about which of the two comparable buildings has the better-supported behaviour: supports of 0.455 and 0.978 at the unit stratum against 0.437 and 0.014 at the device stratum, an inversion. A quantity reproducible by counting on one stratum and reversed on another is not yet a measurement.

5 The conformance null over 73 seeded scenarios

Evaluation used three public LBNL datasets [10, 9] — a single-duct air-handling unit and parallel and series fan-powered units, 74 seeded fault scenarios of which 73 are scored, one series-unit file being excluded for a rotated calendar, and the single-duct system’s 14 being 20 files of which two families of four are byte-identical copies of one run each [18] — scored at day level with the scenario as the unit of analysis [21]. Eight channels score each day: physics rules,

⁴ `outputs/grammar_results.json`, `reversal_probe`.

⁵ `outputs/discovery_predicts_transfer.json`, Q: `sync_days` against the raw occupied-day count.

Table 2. The conformance ablation, read from `outputs/benchmark_v6_*.json`, `meaningful_channels`. “Credited” counts detections in which a conformance channel was among the gated channels; “sole” counts those in which it was the only one. The single-duct count is the naive 14 of 14; one detection is later adjudicated out because the fault-free file was simulated on a different configuration branch from every fault file, which changes no cell because conformance is credited with none.

System	Scenarios	Detected	Credited	Sole
Single-duct AHU	14	14	0	0
Parallel fan-powered	30	26	0	0
Series fan-powered	29	25	7	0
Total	73	65	7	0

temperature-residual bands, *model conformance* and *device conformance* (alignment cost against the discovered unit and device nets — these two, and only these two, are the process-mining channels), absence of a scheduled activity, statistical bands on activity counts and on reversals, and an advisory seasonal rate channel. Every channel is calibrated on the fault-free year only and may report only what exceeds its own measured noise floor (a binomial at $p < 10^{-3}$ against its false-alarm rate on the fault-free holdout days it could evaluate, floored at three days); the conformance channels do not compete against an uncalibrated strawman.

5.1 The ablation

Because the committed scorecards record, per scenario, which gated channels were credited with the detection, the ablation is a read rather than a re-run (Table 2). No scenario on any system is detected only by the two conformance channels: zero sole detections in 73 scenarios bounds the sole-detection rate below 4.1% (one-sided 95%, rule of three), and below 1.5% over the 199 scenarios of all five systems. They are credited with none of the 14 single-duct or 26 parallel-unit detections and appear in 7 of the 25 series-unit detections, never alone (counts after a pre-registered column-coverage audit that left the model and device rows identical and recovered eight then-missed fan-powered scenarios); removing both channels leaves every scorecard unchanged. Nor is removal merely free. On the series unit the two channels appear on six of the nine false-alarm days of the 96-day fault-free holdout (out-of-sample counts), and deleting them eliminates five (the sixth is also flagged by another channel), so that system’s false-alarm rate falls from 9 in 96 to 4 in 96 at no detection cost; on the parallel unit the same deletion takes 8 in 96 to 6.⁶ The process-mining channels’ entire measurable contribution to this detector was five false alarms.

⁶ `outputs/union_fpr_sfpu.json` (after X25): model channel {2018-09-24, 2018-10-31, 2018-11-23, 2018-11-29, 2018-12-25}, device channel {2018-12-28}, frequency channel {2018-11-29}, residual channel {2018-04-27, 2018-06-25, 2018-08-27} after the coverage audit; `union_all18` 9 days.

5.2 The matched-rule control

A second control separates what discovery contributes at design time from what it contributes at run time. For each channel a domain expert hand-wrote the rule the channel’s behaviour implied and scored it on the same scenarios. Two reproduce their channel exactly — 141 against 141 on the series unit and 206 against 206 on the parallel unit, both the frequency channel on the unstable-room-temperature scenario — which is the useful half of the finding: discovery located an invariant nobody had written down, and once written down the channel was exactly redundant with it. The founding hypothesis had a falsifier attached and it fired: on the series unit a one-line rule, *occupied workday and air-handler heating never active*, covered 192 fault days against the model channel’s 135 when the falsifier fired (178 in the current artefact) on the same scenario, with zero false alarms on 365 fault-free days,⁷ and the 135 rests four fifths on a threshold interpolated from one holdout day (22–26 under the most conservative). Transplanted, the same rule fires on 231 fault-free days on the single-duct system, 124 on the parallel unit and none on the series unit; we first read the 231 as evidence about the models and retract it: the single-duct system has no hot-water valve, so *heating never active* fires on every occupied day for want of a sensor. The only informative contrast is 124 against 0, and whether the models predict it is the subject of Section 6.

5.3 Reading the null

Three readings are available. The conformance channels might be poorly implemented; but they use standard alignment-based replay [1] on nets from a standard miner, and a pre-registered positive control (order violations injected into fault-free holdout traces, scored through the deployed gate) shows what the instrument is: on the fan coil and dual-duct units a reversed day is flagged (55 of 72, 59 of 83), so the channel sees order there and the faults do not change it; on the single-duct unit a reversed day is separable on average (AUC 0.91) but fault-free days already fit worse, so no in-budget threshold sees it; on the fan-powered units fitness does not move.⁸ Where the channel can see order it detects no scenario that is not already detected. Nor does another instrument: a pre-registered grid of seven (inductive miner at noise 0.0 and 0.5, token replay, heuristics miner, log skeleton, Declare, temporal profile), each discovered and thresholded as the deployed channel is, detects none of the 33 scenarios it then missed on four systems at their thresholds (realised false-alarm rates 0 to 9.7%; sixteen were later recovered by that audit); two heuristics-miner cells timed out and are reported as not evaluated.⁹ The domain might be wrong for the method; partly, since two of the three nets do not encode order. The third reading: every genuine model-channel detection we inspected was a *count* deviation — extra heating episodes,

⁷ `outputs/matched_rules_sfpu.json`, `SFPU_VAVDMPRStuck_80%: mr1 = 192, model = 178` (135 at the Phase 3B close), `healthy_fp.mr1 = 0`.

⁸ `outputs/x22_positive_control.json`.

⁹ `outputs/x30_method_grid.json`.

collapsed fan cycles, a vanished daily rhythm — surfacing through alignment cost, and once a frequency channel measured count deviations directly and two-sidedly, the conformance channels had nothing uniquely theirs. Alignment cost was acting as a noisy, one-sided, expensive frequency detector; if that generalises, conformance-as-anomaly-detection will underperform wherever the net is order-permissive, which is wherever the alphabet is loop-heavy, as sensor-derived alphabets of complementary pairs cycling many times a day are. Checking the log rather than the model: for eight of the twelve scenarios then missed — all reheat-coil fouling — the state-event log is indistinguishable from healthy at 5% in every daily activity count and state duration, so no method reading that log can detect them at the deployed alphabet; under valve faults it changes in timing and counts, not order. A pre-registered time-infused state model (day-median durations and first-on minutes per state pair) is significant on 11 of 73 scenarios, none the deployed channels miss, at five extra false-alarm days per fan-powered system. Enriching the alphabet with healthy-derived actuator and flow band states (X14) catches one missed scenario at 18 and 21 false-alarm days in 96 on the fan-powered units, over budget; a frequency channel on that alphabet (X15) keeps it in budget but catches none of the fourth system’s ten misses, so the gain does not replicate and is not adopted. The null also transfers: a dual-duct AHU onboarded blind by configuration alone detects 45 of 55 seeded faults at 3 false-alarm days in 96 (50 after the column-coverage audit), the model channel credited on two scenarios other channels already carry and never alone. A fifth, a fan coil unit, detects 40 of 47 at 4 in 96 (43 at 5 after the audit) with the model channel never alone — after a first run whose 25 model-channel false-alarm days (of 28) were all weekends and whose six model-only “detections” were those same weekends; the pre-registered repair moved no earlier scorecard.¹⁰¹¹¹²

6 A pre-registered test that never had power

One claim for the discovered models remained after the null: that they *locate* invariants, so that a rule written from a model’s reading of one building holds on another building where that model’s support for the invariant is high. Three hand-written rules had been transplanted across the three buildings with their fault-free firings recorded, so a test was pre-registered: the model-derived quantity (the fraction of fault-free occupied days whose optimal alignment to the discovered unit net contains a synchronous move on the heating activity), three predictions and three falsifiers were committed before computation; the scorer was changed once afterwards (Monte Carlo to exact permutation p , verdict-neutral, but without a dated amendment); the quantity was computed by an

¹⁰ `outputs/x19_onboarding_run1.json, x19_onboarding.json.`

¹¹ `outputs/x17_onboarding.json.`

¹² `outputs/x12_log_diagnosis.json, x12_time_perspective.json, x14_enriched_alphabet.json, x14_control.json, x15_enriched_frequency.json;` each records its pre-registration and amendments, and X12’s the one prediction that was too weak to discriminate.

agent instructed not to open the rule-firing artefacts. The record, with three dated amendments, is in the repository.¹³

The test failed for four reasons, each caught later than it should have been; the first is the undocumented scorer change, which the second amendment records. The seal was also narrower than intended: the pre-registration’s disclosure section printed the rule’s firings, unblinding the two informative cells. An amendment made before computation excluded the single-duct system for the correct reason given in Section 5, and a second, after the fact, blamed the resulting six-cell design for the test’s lack of power, since two groups of three admit a minimum exact p of 0.05 against a criterion of $p < 0.05$. That diagnosis was itself wrong. The model quantity is a property of a building’s net and is constant across rules within a building (0.000, 0.455 and 0.978); the nine cells were three buildings pseudo-replicated, the valid permutation null is over buildings, and with three the minimum attainable one-sided p is $1/3! = 0.167$; with the two buildings the amended test actually scored it is $1/2! = 0.5$. The test never had power at any stage. The reported $p = 0.20$ is not a measurement, and this paper draws no inference about the claim in either direction.

The lesson is the contribution. Pre-registration fixed the quantity, the predictions and the falsifiers; it did not fix the unit of analysis, and two amendments by the same author under the same discipline did not either. A power calculation on the wrong unit survives pre-registration intact. The remedy is a mandatory pre-registration field naming the unit of replication and the minimum attainable p under a permutation of that unit, computed before any prediction is written; it would have declared this test unpowered at registration. With four scorable buildings the re-run had power and passed ($\rho = -1$, exact $p = 1/24$); but the model quantity is the log’s own heating-day count on three of the four buildings exactly and the transplanted rule is a heating-absence rule, so a discovery-free control added after the result — that count — passes identically. The pass is reported, and it supports nothing about discovery; a net fitted on one building and read against another’s year is not that count either, and, as the pre-registration bound us to report, this first differing ordering runs the wrong way at $n = 4$.¹⁴

7 Tests that pin refutations and disclosures

Every result here is guarded by an automated test that runs on every commit, and what the suite guards is the transferable part. Beyond regenerating results [12] and a byte-reproducibility standard on the single-duct system, it pins three things we have not found guarded elsewhere. It pins *fired falsifiers*: the sampling-rate experiment’s finding is its four refutations, and a regeneration that lost one fails the build as a lost feature would. It pins *disclosure obligations*: that the false-alarm artefact still declares each channel’s threshold provenance, that the post-selection defence for a demoted channel ships beside the result it

¹³ docs/plans/2026-09-23-discovery-predicts-transfer-prereg.md.

¹⁴ outputs/x20_transfer_four.json, x21_foreign_net_support.json;
docs/plans/2026-09-24-x20-transfer-four-buildings-prereg.md.

defends, and that the transfer artefact still states its design’s floor. And it pins *provenance*: the pre-registration, its first amendment and the scorer must be git ancestors of the commit that first added the result. The justification is a pattern the project’s lesson ledger records repeatedly: caveats decay faster than code, so a lesson is not closed until a test enforces it.

8 Limitations

Every detection result comes from simulation (the deployed budget has since been measured at 6.1 % of held-out days on one real rooftop unit), and simulation flatters: noise-free years make thresholds cheap and silence easy. For the null the bias runs the other way — a clean training year should, if anything, help a discovered net — but we cannot exclude that real, noisier logs would change the relative standing of conformance against the statistical channels; on the one LBNL dataset that records the refrigerant side (a simulated rooftop unit) the unchanged detector catches all ten of that unit’s seeded coil-fouling severities at no false alarm in 28 held-out days, condenser fouling through a refrigerant-side residual and evaporator fouling through an air-side one (that dataset seeds fouling as airflow reduction), a positive control for the condenser half of that reading; two pre-registered injections (sensor jitter; field sensor error, change-of-value logging and schedule shifts), run before the coverage audit, left detection counts within three of clean and false alarms at or below 4 in 96.¹⁵¹⁶ Three buildings at the time of the transfer test is the harder limit (a fourth and a fifth were onboarded afterwards; the pre-registered four-building re-run passed at the floor $1/4! = 0.042$, and so did its control): a building is the unit across which a discovered model must transfer, two of the three are variants of one equipment family, and all share a simulated weather-year family. The noise-floor gate applies no correction across roughly 750 binomial tests at $\alpha = 10^{-3}$; per-day flags are autocorrelated through weather, so every p is optimistic; and the conformance thresholds, though calibrated on a separate slice, come from the same fault-free year as every band. Eight machine-verified benchmark defects are reported with reproducing evidence in a companion paper [18]. AI coding agents were used for implementation, analysis and adversarial recomputation (Section 6 reports how far a sealed computation held); all designs, pre-registrations and claims were authored and approved by the author.

9 Conclusion

We carried event-log process mining into a domain it had not been applied to, with a pre-registered hypothesis and a falsifier attached, and the falsifier fired. What survives as a detector contains no discovered model: the process-mining contribution that remains is the event log and the count perspective

¹⁵ `outputs/x31_field_conditions.json`.

¹⁶ `outputs/x9_x10_robustness.json`.

on it. Discovery earned its place at design time, by locating an invariant an expert could then write down, and lost it at run time, exactly and measurably, to the rule it had suggested. The three supporting measurements — a 3.5-point circularity bound, a reversal probe that exposed two nets with fitness above 0.90 as order-blind, and an alignment support that reproduced a raw log count exactly — are cheap, none is specific to buildings, and each should be measured elsewhere.

Acknowledgements. The author thanks Dr. Anthony Aighobahi and Dr. Mridula Sharma for supervision. AI coding assistants were used as stated in Section 8; all designs, pre-registrations and claims were authored and approved by the author.

References

1. van der Aalst, W.M.P., Adriansyah, A., van Dongen, B.: Replaying history on process models for conformance checking and performance analysis. *WIREs Data Mining and Knowledge Discovery* **2**(2), 182–192 (2012). <https://doi.org/10.1002/widm.1045>
2. Berti, A., van Zelst, S., Schuster, D.: PM4Py: A process mining library for Python. *Software Impacts* **17**, 100556 (2023). <https://doi.org/10.1016/j.simpa.2023.100556>
3. Bezerra, F., Wainer, J.: Algorithms for anomaly detection of traces in logs of process aware information systems. *Information Systems* **38**(1), 33–44 (2013). <https://doi.org/10.1016/j.is.2012.04.004>
4. Bi, J., Wang, H., Yan, E., Wang, C., Yan, K., Jiang, L., Yang, B.: AI in HVAC fault detection and diagnosis: A systematic review. *Energy Reviews* **3**(2), 100071 (2024). <https://doi.org/10.1016/j.enrev.2024.100071>
5. vanden Broucke, S.K.L.M., De Weerd, J., Vanthienen, J., Baesens, B.: Determining process model precision and generalization with weighted artificial negative events. *IEEE Transactions on Knowledge and Data Engineering* **26**(8), 1877–1889 (2014). <https://doi.org/10.1109/TKDE.2013.130>
6. Brzychczy, E., Aleknytyte-Resch, M., Janssen, D., Koschmider, A.: Process mining on sensor data: a review of related works. *Knowledge and Information Systems* **67**, 4915–4948 (2025). <https://doi.org/10.1007/s10115-024-02297-y>
7. Crowe, E., Chen, Y., Reeve, H., Yuill, D., Ebrahimifakhar, A., Chen, Y., Troup, L., Smith, A., Granderson, J.: Empirical analysis of the prevalence of HVAC faults in commercial buildings. *Science and Technology for the Built Environment* **29**(10), 1042–1057 (2023). <https://doi.org/10.1080/23744731.2023.2263324>
8. van Dongen, B.F., Carmona, J., Chatain, T.: A unified approach for measuring precision and generalization based on anti-alignments. In: *Business Process Management (BPM 2016)*. LNCS, vol. 9850, pp. 39–56. Springer (2016). https://doi.org/10.1007/978-3-319-45348-4_3
9. Granderson, J., Lin, G., Chen, Y., Casillas, A., Wen, J., Chen, Z., Im, P., Huang, S., Ling, J.: A labeled dataset for building HVAC systems operating in faulted and fault-free states. *Scientific Data* **10**, 342 (2023). <https://doi.org/10.1038/s41597-023-02197-w>
10. Granderson, J., Lin, G., Harding, A., Im, P., Chen, Y.: Building fault detection data to aid diagnostic algorithm creation and performance testing. *Scientific Data* **7**, 65 (2020). <https://doi.org/10.1038/s41597-020-0398-6>

11. Janssen, D., Mannhardt, F., Koschmider, A., van Zelst, S.J.: Process model discovery from sensor event data. In: *Process Mining Workshops (ICPM 2020)*. LNBIP, vol. 406, pp. 69–81. Springer (2021). https://doi.org/10.1007/978-3-030-72693-5_6
12. Jimenez, I., Sevilla, M., Watkins, N., Maltzahn, C., Lofstead, J., Mohror, K., Arpaci-Dusseau, A., Arpaci-Dusseau, R.: The Popper convention: Making reproducible systems evaluation practical. In: *IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)* (2017). <https://doi.org/10.1109/IPDPSW.2017.157>
13. Leemans, S.J.J., Fahland, D., van der Aalst, W.M.P.: Discovering block-structured process models from event logs — a constructive approach. In: *Application and Theory of Petri Nets and Concurrency (PETRI NETS 2013)*. Lecture Notes in Computer Science, vol. 7927, pp. 311–329. Springer (2013). https://doi.org/10.1007/978-3-642-38697-8_17
14. Muñoz-Gama, J., Carmona, J.: A fresh look at precision in process conformance. In: *Business Process Management (BPM 2010)*. Lecture Notes in Computer Science, vol. 6336, pp. 211–226. Springer (2010). https://doi.org/10.1007/978-3-642-15618-2_16
15. Nolle, T., Luetzgen, S., Seeliger, A., Mühlhäuser, M.: BINet: Multi-perspective business process anomaly classification. *Information Systems* **103**, 101458 (2022). <https://doi.org/10.1016/j.is.2019.101458>
16. Pitsch, C., Brockhoff, T., Adams, J.N., Leemans, S.J.J., Celi, L.A., van der Aalst, W.M.P.: Hypothesis testing for processes. In: *2025 7th International Conference on Process Mining (ICPM)*. pp. 1–8. IEEE (2025). <https://doi.org/10.1109/ICPM66919.2025.11220677>
17. Rozinat, A., van der Aalst, W.M.P.: Conformance checking of processes based on monitoring real behavior. *Information Systems* **33**(1), 64–95 (2008). <https://doi.org/10.1016/j.is.2007.07.001>
18. Singh, Y.: Five defects in the LBNL HVAC fault-detection benchmarks, and the gate battery that found them (2027), companion paper, submitted to the R2PM workshop at ICPM 2027; in preparation
19. Tax, N., Lu, X., Sidorova, N., Fahland, D., van der Aalst, W.M.P.: The imprecisions of precision measures in process mining. *Information Processing Letters* **135**, 1–8 (2018). <https://doi.org/10.1016/j.ipl.2018.01.013>
20. Weytjens, H., De Weerd, J.: Creating unbiased public benchmark datasets with data leakage prevention for predictive process monitoring. In: *Business Process Management Workshops (BPM 2021)*. Lecture Notes in Business Information Processing, vol. 436, pp. 18–29. Springer (2022). https://doi.org/10.1007/978-3-030-94343-1_2
21. Yuill, D.P., Braun, J.E.: Evaluating the performance of fault detection and diagnostics protocols applied to air-cooled unitary air-conditioning equipment. *HVAC&R Research* **19**(7), 882–891 (2013). <https://doi.org/10.1080/10789669.2013.808135>
22. van Zelst, S.J., Mannhardt, F., de Leoni, M., Koschmider, A.: Event abstraction in process mining: literature review and taxonomy. *Granular Computing* **6**(3), 719–736 (2021). <https://doi.org/10.1007/s41066-020-00226-2>
23. Zhong, Y., Lisitsa, A.: Can process mining help in anomaly-based intrusion detection? arXiv:2206.10379 (2022), preprint; not peer reviewed. Reports that naïve process mining on network event logs was ineffective for intrusion detection